

Chengyuan ZHANG

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RESEARCH INTERESTS

My research develops stochastic models of human behavior and the Bayesian inference that fits them to data. Deterministic behavioral models discard whatever context cannot explain, but naturalistic data show that this residual is structured: it carries memory, it differs systematically between individuals, and it switches between latent behavioral regimes. Modeling that structure explicitly is what allows uncertainty to travel from individual decisions through to the collective dynamics they produce, and it is what makes a traffic simulation credible enough to test decisions on.

- Stochastic behavior modeling: memory, individual heterogeneity, and latent regimes in human decision sequences. [[ISTTT26](#); [TR Part C '24](#); [IEEE T-ITS '24](#)]
- Bayesian inference and calibration at scale: hierarchical models, simulation-based inference, online estimation, and benchmarks for judging calibration quality. [[arXiv:2602.05246](#); [IEEE IV '26](#); [AutoTune benchmark](#); [IEEE T-ITS '24](#)]
- Structure learning in spatiotemporal systems: latent low-rank and nonstationary structure, from individual trajectories to city-scale road networks. [[IEEE TKDE '24](#); [Transportation Science '25](#)]
- Multi-agent interaction: representing how road users respond to one another, and testing when that response actually matters. [[IEEE T-ITS '22](#); [IEEE ITSC '23](#); [Foundations and Trends in Robotics '22](#)]

Directions I am building toward: interaction between people and automated vehicles, pedestrian behavior, and world models with enough cognitive structure to explain why someone acted rather than only what they did next.

EDUCATION

McGill University

Montréal, Québec, Canada

Ph.D. in Civil Engineering, supervised by Prof. Lijun Sun

Sept. 2020 – May 2026

- **Dissertation:** “From Micro Interactions to Traffic Flow: Stochastic Driver Models for Realistic Traffic Simulation”
- Built stochastic extensions of the intelligent driver model that capture correlated errors, driver-level heterogeneity and latent regime switching; calibrated them on naturalistic highway trajectories; and carried the resulting parameter uncertainty through multi-vehicle simulation to the traffic flow that emerges from it.
- Fast tracked to the Ph.D. program in 2022 winter with 4.00/4.00 Master’s GPA;

Chongqing University

Chongqing, China

B.Eng. in Vehicle Engineering

Sept. 2015 – June 2019

- **Thesis:** “Recognition of Multi-Vehicle Interaction Patterns Based on Probabilistic Velocity Fields”
- Selected to **Excellent Student Program** (top 5%, on basis of outstanding academic performance;)
- Completed extensive coursework in **Electronics Science and Technology** (2015-2016.)

WORK AND VISITING RESEARCH EXPERIENCE

Postdoctoral Researcher

Montréal, Québec, Canada

Department of Civil Engineering, McGill University

May 2026 – Present

Visiting Student Researcher

Pittsburgh, Pennsylvania, USA

Robotics Institute, Carnegie Mellon University

Jan. 2023 – Aug. 2023

Visiting Student Researcher

Berkeley, California, USA

Department of Mechanical Engineering, UC Berkeley

Sept. 2019 – Jan. 2020

Visiting Student Researcher

Pittsburgh, Pennsylvania, USA

Department of Mechanical Engineering, Carnegie Mellon University

July 2018 – Oct. 2018

Visiting Student

Philadelphia, Pennsylvania, USA

University of Pennsylvania

Jan. 2018 – Feb. 2018

PREPRINTS AND MANUSCRIPTS UNDER REVIEW

Author notation: **bold** indicates the author of this CV; * corresponding author; † equal contribution.

- **Chengyuan Zhang**, Zhengbing He, Cathy Wu, and Lijun Sun*, “When context is not enough: Modeling unexplained variability in car-following behavior,” *Transportation Research Part B: Methodological*, 2025, Under review. Accepted for podium presentation at the 26th International Symposium on Transportation and Traffic Theory (ISTTT26). arXiv: [2507.07012](#). [[arXiv](#)]
- Menglin Kong, **Chengyuan Zhang***, and Lijun Sun, “Active simulation-based inference for scalable car-following model calibration,” *arXiv preprint arXiv:2602.05246*, 2026. [[arXiv](#)]
- **Chengyuan Zhang**, Cathy Wu, and Lijun Sun*, “Markov regime-switching intelligent driver model for interpretable car-following behavior,” *arXiv preprint arXiv:2506.14762*, 2025. [[arXiv](#)]

JOURNAL PUBLICATIONS

- Xinyu Chen, **Chengyuan Zhang**, Xi-Le Zhao, Nicolas Saunier*, and Lijun Sun, “Forecasting sparse movement speed of urban road networks with nonstationary temporal matrix factorization,” *Transportation Science*, vol. 59, no. 3, 670–687, 2025. [[TranSci](#)] [[arXiv](#)] [[slides](#)] [[Github](#)] [[blog](#)]
- **Chengyuan Zhang**, Wenshuo Wang, and Lijun Sun*, “Calibrating car-following models via Bayesian dynamic regression,” *Transportation Research Part C: Emerging Technologies*, vol. 168, 104719, 2024. [[TR PartC](#)] [[arXiv](#)] [[Github](#)]
- **Chengyuan Zhang** and Lijun Sun*, “Bayesian calibration of the intelligent driver model,” *IEEE Transactions on Intelligent Transportation Systems*, vol. 25, no. 8, 9308–9320, 2024. [[IEEE TITS](#)][[arXiv](#)] [[Github](#)] [[presentation](#)] [[poster](#)]
- Xiaoxu Chen, **Chengyuan Zhang**, Zhanhong Cheng, Yuang Hou, and Lijun Sun*, “A Bayesian Gaussian mixture model for probabilistic modeling of car-following behaviors,” *IEEE Transactions on Intelligent Transportation Systems*, vol. 25, no. 6, 5880–5891, 2024. [[IEEE TITS](#)]
- Xinyu Chen†, **Chengyuan Zhang**†, Xiaoxu Chen, Nicolas Saunier, and Lijun Sun*, “Discovering dynamic patterns from spatiotemporal data with time-varying low-rank autoregression,” *IEEE Transactions on Knowledge and Data Engineering*, vol. 36, no. 2, 504–517, 2024. [[IEEE TKDE](#)] [[arXiv](#)] [[data & code](#)] [[blog](#)]
- **Chengyuan Zhang**, Jiacheng Zhu, Wenshuo Wang*, and Junqiang Xi, “Spatiotemporal learning of multivehicle interaction patterns in lane-change scenarios,” *IEEE Transactions on Intelligent Transportation Systems*, vol. 23, no. 7, 6446–6459, 2022. [[demo](#)] [[IEEE TITS](#)] [[project website](#)] [[Github](#)]
- **Chengyuan Zhang**, Xiaomin Zhang, Hongyun Ye, Ming Wei, and Xianxiong Ning*, “An efficient parking solution: A cam-linkage double-parallellogram mechanism based 1-degrees of freedom stack parking system,” *Journal of Mechanisms and Robotics*, vol. 11, no. 4, 045001, 2019. [[demo](#)] [[ASME JMR](#)]
- **Chengyuan Zhang** and Jian Xiao*, “Chaotic behavior and feedback control of magnetorheological suspension system with fractional-order derivative,” *Journal of Computational and Nonlinear Dynamics*, vol. 13, no. 2, 021007, 2018. [[ASME JCND](#)]

INVITED MONOGRAPH AND REVIEW

- Wenshuo Wang, Letian Wang, **Chengyuan Zhang**, Changliu Liu, and Lijun Sun, “Social interactions for autonomous driving: A review and perspectives,” *Foundations and Trends® in Robotics*, vol. 10, no. 3-4, 198–376, 2022. [[ebook](#)] [[arXiv](#)]

CONFERENCE PUBLICATIONS

- Menglin Kong, **Chengyuan Zhang**, and Lijun Sun^{*}, “Online calibration of context-driven car-following models,” in *2026 IEEE Intelligent Vehicles Symposium (IV)*, To appear, IEEE, 2026.
- Cameron Hickert, Athena Wang, Maryam Samaei, **Chengyuan Zhang**, Lijun Sun, Yanbing Wang, Mostafa Ameli, and Cathy Wu^{*}, “AutoTune: A unified benchmark for highway traffic microsimulation calibration,” in *2026 IEEE Intelligent Vehicles Symposium (IV)*, To appear, IEEE, 2026.
- **Chengyuan Zhang**[†], Kehua Chen[†], Meixin Zhu, Hai Yang, and Lijun Sun^{*}, “Learning car-following behaviors using Bayesian matrix normal mixture regression,” in *2024 IEEE Intelligent Vehicles Symposium (IV)*, IEEE, 2024, 608–613. [[arXiv](#)][[IEEE IV](#)]
- **Chengyuan Zhang**, Rui Chen, Jiacheng Zhu, Wenshuo Wang, Changliu Liu, and Lijun Sun^{*}, “Interactive car-following: Matters but NOT always,” in *2023 IEEE 26th International Conference on Intelligent Transportation Systems (ITSC)*, IEEE, 2023, 5120–5125. [[arXiv](#)][[IEEE ITSC](#)]
- Wenshuo Wang, **Chengyuan Zhang**, Pin Wang, and Ching-Yao Chan^{*}, “Learning representations for multi-vehicle spatiotemporal interactions with semi-stochastic potential fields,” in *2020 IEEE Intelligent Vehicles Symposium (IV)*, IEEE, 2020, 1935–1940. [[IEEE IV](#)]
- **Chengyuan Zhang**, Jiacheng Zhu, Wenshuo Wang, and Ding Zhao^{*}, “A general framework of learning multi-vehicle interaction patterns from video,” in *2019 IEEE Intelligent Transportation Systems Conference (ITSC)*, IEEE, 2019, 4323–4328. [[IEEE ITSC](#)]

BOOKS

- Xinyu Chen, Jieling Jin, Qionghua Liao, **Chengyuan Zhang**, and Xiaoxu Chen, *Academic Writing with LaTeX (in Chinese)*. 2023. [[ebook](#)] [[link](#)]

PATENT

- [CN108222589B](#), Cam-connecting rod type mechanical three-dimensional parking device. Chengyuan Zhang, Xiaomin Zhang, Hongyun Ye, Jinming Shi, Manzhi Wang, and Xianxiong Ning. *Chongqing University*.

RESEARCH GRANTS

- “Quantitative Evaluation and Modeling of Action-Reaction Cycles in Interactive Human Driving Behaviors”, NSERC Alliance International Catalyst Grant, \$25,000, 2022-2023. Drafted the majority of the proposal; named trainee (PI: Prof. Lijun Sun).

INVITED TALKS AND SEMINARS

- Beyond White Noise: What Naturalistic Data Reveal About Memory, Heterogeneity, and Latent Regimes in Car-Following. Chair of Econometrics and Statistics, TU Dresden, Dresden, Germany. July 2026. *Host: Dr. Martin Treiber*.
- Discovering Urban Mobility Patterns from Human Behavior to City-Scale Dynamics with Interpretable AI. MIT Senseable City Lab, Massachusetts Institute of Technology. Apr. 2026. *Host: Prof. Carlo Ratti*.
- Modeling Human Driving as a Dynamic Interaction Process: Uncertainty-Aware Learning from Naturalistic Driving Data. MIT AgeLab, Massachusetts Institute of Technology. Apr. 2026. *Host: Dr. Pnina Gershon*.
- From Micro Interactions to Traffic Flow: Stochastic Driver Models for Realistic Traffic Simulation. TOPS Lab, Department of Transportation Engineering, Tongji University, Shanghai. Feb. 2026. *Host: Prof. Jian Sun*.
- From Micro Interactions to Traffic Flow: Stochastic Driver Models for Realistic Traffic Simulation. MIT Wu Lab, Massachusetts Institute of Technology. Oct. 2025. *Host: Prof. Cathy Wu*.
- Stochastic Modeling and Simulations of Car-Following Behaviors. JTL Urban Mobility Lab, Massachusetts Institute of Technology. Feb. 2025. *Host: Prof. Xinyu Chen*.
- Bayesian Calibration of Car-Following Models. Department of Mechanical Engineering, Beijing Institute of Technology, Beijing. Nov. 2024. *Host: Prof. Wenshuo Wang*.
- Bayesian Calibration of Car-Following Models. Artificial Intelligence Institute, Shanghai Jiao Tong University, Shanghai. Oct. 2024. *Host: Prof. Yanyan Xu*.

- Bayesian Calibration of the IDM. General webinar series of the TRB Standing Committee on Traffic Flow Theory and Characteristics (ACP50). Feb. 2023. *Host: Dr. Garyoung Lee.*
- Learning to Drive: Imitating How Human Drivers Make Decisions in Complicated Urban Scenarios. Department of Mechanical Engineering, UC Berkeley. Dec. 2019. *Host: Prof. Masayoshi Tomizuka.*
- Design of A Novel Cam-Linkage Mechanism. The 1st Workshop for SAILER at CMU, Department of Mechanical Engineering, Carnegie Mellon University, Pittsburgh. Aug. 2018. *Host: Prof. Ding Zhao.*

CONFERENCE PRESENTATIONS

- **Podium.** When Context Is Not Enough: Modeling Unexplained Variability in Car-Following Behavior. The 26th International Symposium on Transportation and Traffic Theory (ISTTT26). Technical University of Munich, Munich, Germany. July 2026.
- **Podium.** Calibrating Car-Following Models via Bayesian Dynamic Regression. The 25th International Symposium on Transportation and Traffic Theory (ISTTT25). University of Michigan, Ann Arbor. July 2024.
- Poster. When context is not enough: Modeling unexplained variability in car-following behavior. Transportation Research Board 105th Annual Meeting. Washington, DC. Jan. 2026.
- Poster. Markov regime-switching intelligent driver model for interpretable car-following behavior. Transportation Research Board 105th Annual Meeting. Washington, DC. Jan. 2026.
- Poster. Bayesian Calibration of the IDM. Transportation Research Board Annual Meeting. Washington, DC. Jan. 2023.

TEACHING

Traffic Engineering and Simulation (by Prof. Lijun Sun) *2022 Fall, 2023 Fall, 2025 Fall*
Teaching Assistant, Department of Civil Engineering, McGill University

- TA Tutorials: traffic flow fundamentals and simulation; fundamental diagrams and shockwaves.

Basics of Machine Learning and Data Analysis (by Prof. Pradeep Ravikumar) *2022 Fall*
Teaching Assistant (Online), Department of Computer Science, Carnegie Mellon University

- TA Tutorials: ordinary/generalized least squares (OLS/GLS), logistic regression, Bayesian inference, Gaussian processes.

Statistical Machine Learning (by Prof. Dino Sejdinovic) *2020 Spring, 2020 Summer, and 2022 Summer*
Teaching Assistant (Online), Department of Statistics, University of Oxford

- TA Tutorials: LSE, MLE, Linear Regression, PCA, SVM, kernel methods, Bayesian Optimization, Deep Neural Network.

MENTORING

Menglin Kong, Ph.D. student (advised by Prof. Lijun Sun) *2025–2026*
Research supervision, Department of Civil Engineering, McGill University

- Led the project on scalable and online calibration of car-following models; corresponding author on the resulting preprint. Outputs: *IEEE IV 2026* (to appear); arXiv:2602.05246.

Yuang Hou, undergraduate researcher *2022–2023*
Research supervision, Department of Civil Engineering, McGill University

- Supervised his research on probabilistic modelling of car-following behavior; co-author on *IEEE Transactions on Intelligent Transportation Systems* (2024). Subsequently M.S. at Carnegie Mellon University.

LEADERSHIP AND SERVICE

- (Co-)Chair/(Co-)Organizer:
 - 2nd SIAM workshop on IEEE IV24', Jeju Shinhwa World, Jeju Island, Korea, 2024. [[Website](#)]
 - 1st SIAM workshop on IEEE IV23', Anchorage, Alaska, USA, 2023. [[Website](#)]
- Journal Reviewer (22 journals):

- Transportation Science;
- Transportation Research Part B: Methodological;
- Transportation Research Part C: Emerging Technologies;
- European Journal of Operational Research;
- IEEE Transactions on Intelligent Transportation Systems;
- IEEE Transactions on Intelligent Vehicles;
- IEEE Transactions on Cybernetics;
- IEEE Transactions on Industrial Electronics;
- Artificial Intelligence for Transportation;
- Transportation Research Interdisciplinary Perspectives;
- Travel Behaviour and Society;
- Journal of Traffic and Transportation Engineering;
- Advanced Engineering Informatics;
- Automotive Innovation;
- ASME Journal of Mechanism and Robotics;
- ASME Journal of Mechanism Design;
- Physica A: Statistical Mechanics and its Applications;
- Nonlinear Dynamics;
- IEEE Transactions on Consumer Electronics;
- IEEE Sensors Journal;
- Scientific Reports;
- Journal of King Saud University Computer and Information Sciences;
- Conference Reviewer:
 - IEEE International Conference on Intelligent Transportation Systems (ITSC), 22-24;
 - IEEE Intelligent Vehicles Symposium (IV), 23-24;
 - Symposium on Machine Learning for Autonomous Driving; [[NeurIPS'23](#)] [[AAAI'25](#)]
 - Transportation Research Board Annual Meeting;
 - The 5th Symposium on Management of Future Motorway and Urban Traffic Systems (MFTS), 2024;
 - RSS 2024 Workshop on Autonomous Vehicles Across Scales (AVAS), 2024; [[RSS'24](#)]
 - Bayesian Decision-making and Uncertainty Workshop; [[NeurIPS'24](#)]
 - International Symposium on Transportation Data & Modelling (ISTDM), 2025;
- Memberships: IEEE (Graduate Student) and IEEE Intelligent Transportation Systems Society (ITSS); Chinese Overseas Transportation Association (COTA); Interuniversity Research Centre on Enterprise Networks, Logistics and Transportation (CIRRELT); Society of Automotive Engineers (SAE) of China;

FELLOWSHIPS AND SCHOLARSHIPS

Over CAD \$300K in competitively awarded fellowships and scholarships.

- McGill Engineering Doctoral Award (MEDA), CAD \$111,000, McGill University, 2022-2025
- FRQNT Doctoral Scholarship (B2X), CAD \$100,000, 2023-2027. “Évaluation quantitative et modélisation des cycles action-réaction dans les comportements de conduite interactifs” [[Link](#)]
- IVADO Excellence Scholarship (M.Sc.), CAD \$40,000, 2020-2022. “Statistical Modeling Framework to Understand Dynamic Traffic Patterns from Video Data”
- FRQNT Master’s Scholarship (B1X), CAD \$17,500, 2022-2023. “Étalonnage bayésien des modèles de suivi de voiture” [[Link](#)]
- CIRRELT Doctoral Scholarship of Excellence, Final Stages, CAD \$7,500, 2025 [[Link](#)]
- CIRRELT Doctoral Scholarship of Excellence, Early Stages, CAD \$4,000, 2023 [[Link](#)]
- CIRRELT Master’s Scholarship of Excellence, CAD \$3,500, 2022
- Graduate Excellence Fellowship Awards, McGill University, 2020

AWARDS AND RECOGNITION

- Honorable Mention, 2024 Clifford Spiegelman Student Paper Competition, Transportation Statistics Interest Group, American Statistical Association, 2024 [[Link](#)]
- Ron Rice Memorial Award, McGill University, 2023
- Mitacs Globalink Research Award - Abroad, 2022-2023
- Graduate Mobility Award, McGill University, 2022-2023

- Graduate Research Enhancement and Travel Awards, McGill University, 2023

EARLIER AWARDS (UNDERGRADUATE)

- Outstanding Undergraduate Thesis Award, Chongqing University, 2019
- Outstanding Graduate of Chongqing University, 2019
- Outstanding Team Award & Best Application Award at DeeCamp, China, 2019
- First Prize in the 2018 National College Mechanical Innovation Competition (TOP #10 in China), 2018
- Outstanding Student Award, Chongqing University (top 3%), 2017 and 2018
- National Scholarship, China, 2017
- First Prize in the Chongqing University Physics Contest (#1/500 in Chongqing University), 2015

OPEN SCIENCE / SOFTWARE / ADDITIONAL OUTPUTS

- https://github.com/Chengyuan-Zhang/IDM_Bayesian_Calibration;
- https://github.com/Chengyuan-Zhang/Gaussian_Velocity_Field;
- Simulator: <https://chengyuan-zhang.github.io/traffic-sim/>;
- Research Notes: <https://chengyuan-zhang.github.io/notes/>;